

Database Normalization

Normalization is the process of organizing data in a database to reduce redundancy and improve data integrity. It was introduced by Edgar F. Codd in 1970 as part of the relational model.

First Normal Form

A relation is in first normal form (1NF) if every attribute contains only atomic values. Repeating groups are not permitted. This is the minimum requirement for a relational table.

Second Normal Form

A relation is in second normal form (2NF) if it is in 1NF and every non-prime attribute is fully functionally dependent on the whole of every candidate key. Partial dependencies are eliminated at this stage.

Third Normal Form

A relation is in third normal form (3NF) if it is in 2NF and no non-prime attribute is transitively dependent on any candidate key. Transitive dependency means an attribute depends on another non-prime attribute rather than directly on the key.

The Boyce-Codd normal form (BCNF) is a stricter version of 3NF. Every determinant must be a candidate key. Most relations in 3NF are also in BCNF, but exceptions exist when a relation has multiple overlapping candidate keys.

Denormalization

Denormalization deliberately introduces redundancy to improve read performance. It trades write cost and integrity risk for query speed, and is applied only after measurement shows a real bottleneck.

Normal Form	Eliminates	Requires
1NF	Repeating groups	Atomic values
2NF	Partial dependency	1NF
3NF	Transitive dependency	2NF